

Oriel Summer school

Secular stagnation and science

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August 2024



Outline

1. Growth, productivity and innovation
2. Secular stagnation
3. The return on research

Innovation and long-run growth

- Economic growth is primarily driven by productivity growth.
- Productivity growth, in the long run, is driven by technological improvement.
- We will illustrate this using two models:
 - Solow Growth Model
 - Romer's Endogenous Growth Model

Why should we care

- Productivity improvement allows us to do more with less
- Allows for leisure, culture, health, education retirement etc...
- Without productivity improvement, (almost) all of us will be illiterate peasants working the same fields our ancestors worked on
- Innovation $\mapsto$ Productivity $\mapsto$ Growth $\mapsto$ Better life! (not always)

Solow Growth Model: Overview

- The Solow model emphasizes capital accumulation, labour
- Production function:

$$Y = AK^{\alpha}L^{1-\alpha}$$

- Y : Output, K : Capital, L : Labor, A : Technology (TFP)
- Technological progress (A) is exogenous.

Solow Growth Model: Role of Technology

- The marginal product of capital is given by

$$\frac{\alpha A}{\left(\frac{K_t}{L_t}\right)^{1-\alpha}}$$

- Long-term growth rate of output per worker ($y = Y/L$):

$$\frac{\dot{y}}{y} = \frac{\dot{A}}{A}$$

- In the Solow model, technological progress ($\dot{A}/A$) is the only source of sustained growth in y .

Decreasing Marginal Returns to Capital at work!

- Adding more of a factor of production (like monitors) initially increases productivity.
- However, the marginal gain decreases as more units are added.
- Until you introduce a new technology!
- Like monitors and VR

Romer's Endogenous Growth Model

- In the Solow model, technological change comes from heaven. Romer changes that
- Output Function:

$$Y_t = K_t^\alpha (A_t L_{Yt})^{1-\alpha}$$

- Technology Growth:

$$\dot{A}_t = A_t L_{At}$$

- Labor Allocation:

$$L_t = L_{Yt} + L_{At}$$

Resource Allocation and Implications

- Resources are divided between good production and new knowledge (RD).
- Implications:
 - Sustained Economic Growth: Continuous investment in RD and human capital.
 - Role of Policy: Supportive policies for education, innovation, and IP rights enhance growth.
 - Knowledge Spillovers: Policies that enhance the diffusion of knowledge to boost economic growth.

Empirics

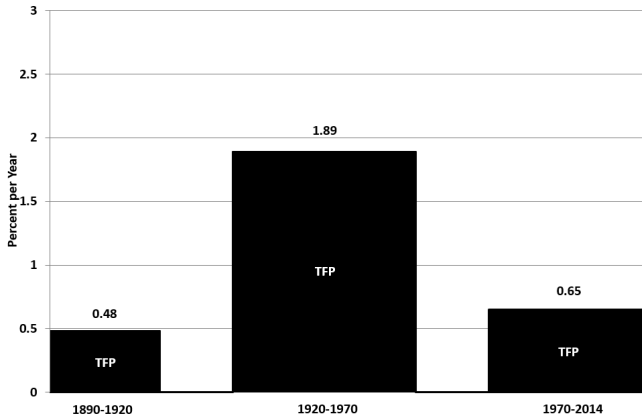
- Recent discussions about economic growth highlight concerns about secular stagnation.
- Evidence shows a decline in productivity growth rates over recent decades, despite a large increase in the research population, contradicting Romer's insight.
- Will growth be sustainable in the future?

Three Eras of Productivity Growth

- Productivity growth has varied significantly across different eras.
- Major productivity gains were observed in 1930-1970.
- The third industrial revolution has not produced similar gains yet

Three eras of productivity Growth

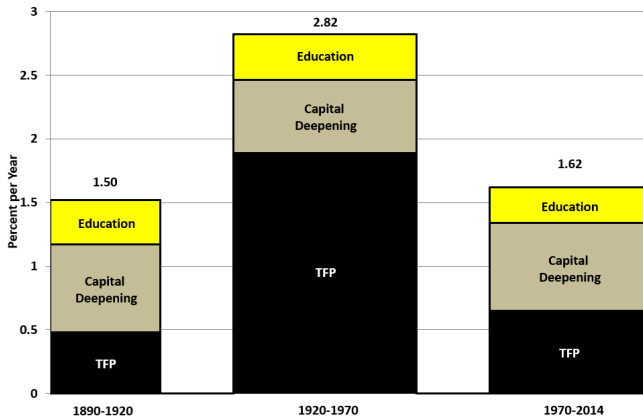
Figure 1-2. Average Annual Growth Rates of Total Factor Productivity, Selected Intervals, 1890-2014



Source: R.Gordon, the rise and fall of American growth

Three eras of productivity Growth

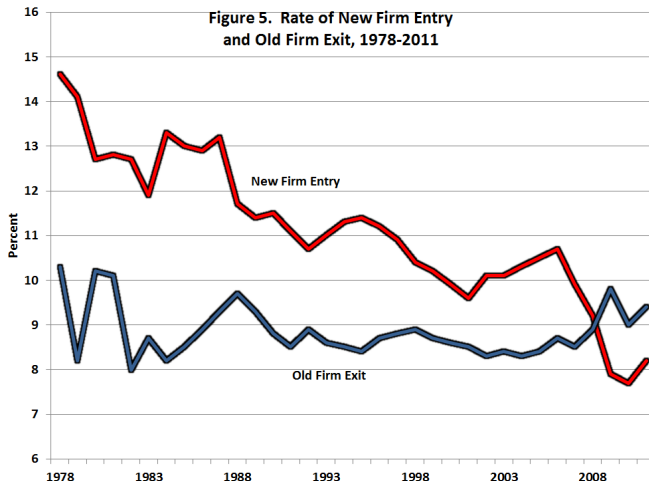
Figure 1-2. Average Annual Growth Rates of Output per Hour and Its Components, Selected Intervals, 1890-2014



Source: R.Gordon, the rise and fall of American growth

Other signs of slow down

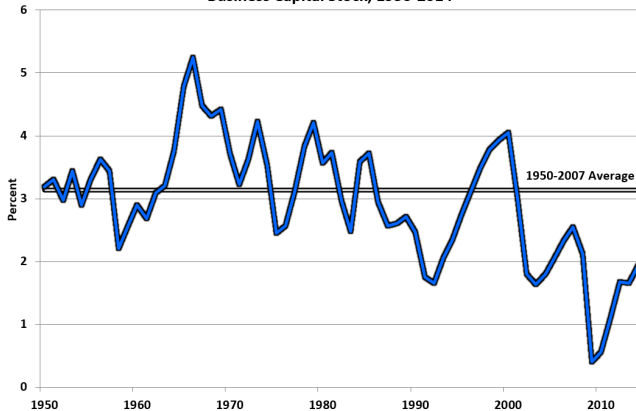
Business dynamism has been continuously declining in the US since the 70s (again from R.Gordon's book)



Other signs of slow down

Business dynamism has been continuously declining in the US since the 70s (again from R.Gordon's book)

Figure 17-6. Annual Ratio of Net Private Business Investment to Private Business Capital Stock, 1950-2014



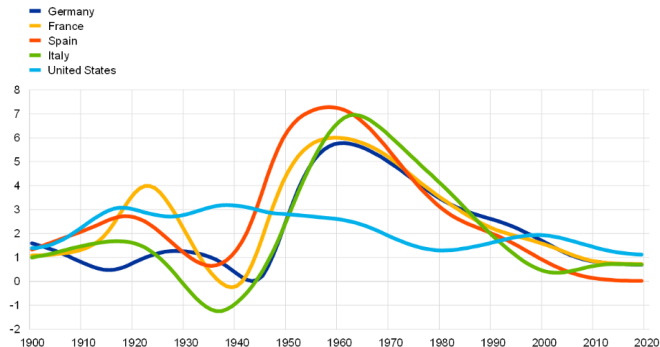
Sources: BEA Fixed Assets Accounts, Tables 4.1, 4.4, and 4.7.

Other countries

Chart 1

Trends in labour productivity (GDP per hour worked) growth in selected euro area countries and the United States

(smoothed annual percentage change)

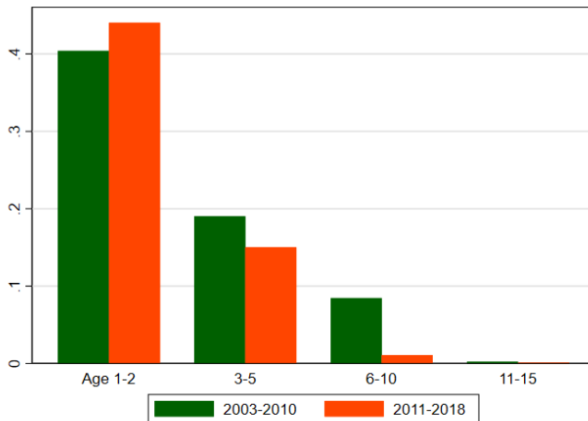


Sources: Own calculations based on Bergeaud, A., Cetto, G. and Lecat, R., "Productivity Trends in Advanced Countries between 1890 and 2012", *Review of Income and Wealth*, Vol. 62, No 3, 2016, pp. 420-444.

Note: The trend is calculated using a Hodrick-Prescott filter with a smoothing parameter λ of 10.

What about China?

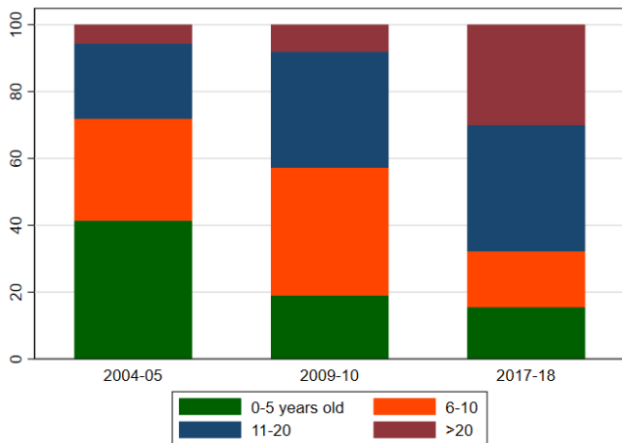
Figure 4: 3-year growth rate of revenues



Notes: This figure plots the 3-year growth rate of revenues for firms of different ages, relative to firms in the 15-year average European market.

What about China?

Figure 3: Revenue share of firms by age group



The return on research

- Romer's model assumes that the production of ideas scales linearly with the number of researchers.
- This leads to increasing returns to scale due to the non-rivalry of ideas.
- Recall that the δ in the Romer model determines the relationship between growth and the number of researchers.
- Recent research shows that this delta has been continuously falling

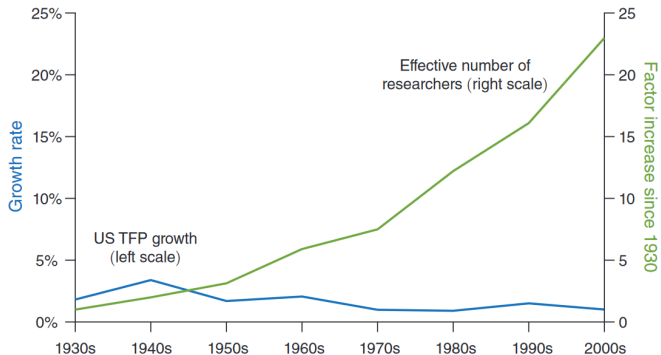
Empirical estimation

- Various empirical studies have challenged Romer's assumption.
- Bloom et al. (2020) documented a sharp decrease in returns to RD in many sectors.
- Large publicly traded companies allocate an ever greater share of human and physical capital to maintain the same growth rate.

The productivity of researchers

VO. 4

BLOOM ET AL.: ARE IDEAS GETTING HARDER TO FIND?



Implications of Decreasing Returns to RD

- Decreasing returns to RD imply that more resources are needed for the same level of technological progress.
- This could lead to slower economic growth if the productivity of RD continues to decline.

One explanation: The burden of knowledge

As knowledge accumulates, researchers must spend more time learning existing knowledge before contributing new innovations. This leads to:

- Longer education and training periods for researchers.
- Higher specialization within fields.
- Slower rate of discoveries and technological advancements.

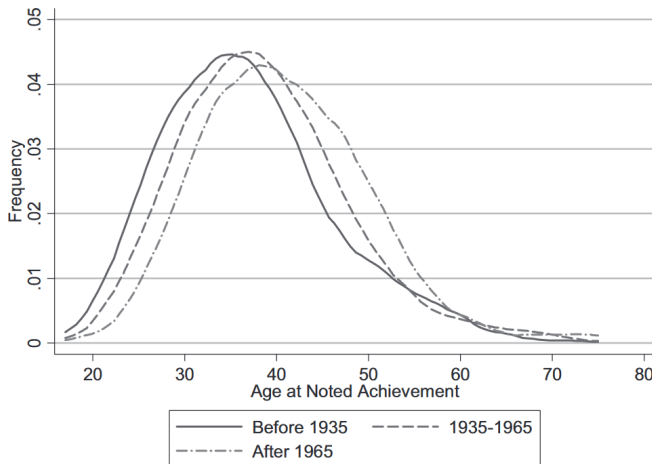
One explanation: The burden of knowledge

Consequences:

- Diminishing returns to research and development (RD) efforts.
- Increased collaboration and teamwork to manage the growing knowledge base.
- Potential slowdown in overall economic growth due to slower productivity growth.

One explanation: The burden of knowledge

FIGURE 2.—SHIFTS IN THE AGE DISTRIBUTION OF GREAT INNOVATION

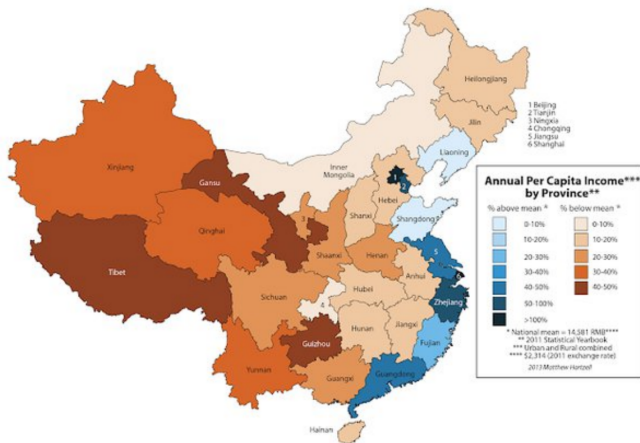


Note: The figure pools the Nobel Prize winner and great inventor data.

Relevant for China?

- China is a middle-income country characterised by very-rich and very poor regions.
- Poor regions are farther from the technological frontier $\mapsto$ No need for innovation to grow $\mapsto$ not affected by secular stagnation in the short run.
- Rich regions are closer to the technological frontier and will be affected in the short run.

Relevant for China?



Larger pool of researcher

China's contributed greatly to the increase in the global population of educated workers. It might help fight off declining research productivity for a while.

That will be your job!